Archisman Panigrahi

Ph.D. Candidate in Physics

Massachusetts Institute of Technology, Cambridge, MA, USA

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Education

Ph.D. in Physics (ongoing) — *Cambridge, MA, USA*

MASSACHUSETTS INSTITUTE OF TECHNOLOGY — *September 2022 - 2027 (expected)*

- C.G.P.A - $5.0/5.0$

Supervisor: Prof. Leonid Levitov

Master of Science in Physics — *Bangalore, India*

INDIAN INSTITUTE OF SCIENCE — *Aug. 2021 - Jun. 2022*

- C.G.P.A - $9.8/10.0$

Bachelor of Science (Research) in Physics — *Bangalore, India*

INDIAN INSTITUTE OF SCIENCE — *Aug. 2017 - Jun. 2021*

- C.G.P.A - $9.8/10.0$

Research Articles

LISTED IN THE CHRONOLOGICAL ORDER OF THEIR APPEARANCE ON ARXIV.

- **A. Panigrahi**, K. Nazaryan; *Viscochiral Transport: Chiral Selection of Hydrodynamic Vortices by Berry Curvature* arXiv:2608.26102
- **A. Panigrahi**, V. Poliakov, L. Levitov; *Particle-Hole Ghost Interference in Superconductors* arXiv:2606.06437
- Y. Liao, A. Grankin, **A. Panigrahi**, V. Galitski, L. Levitov; *Tunable viscosity across the BCS-BEC crossover* arXiv:2604.10759
- **A. Panigrahi**, D. Kaplan; *Valley polarization of chiral excitonic bound states induced by band geometry* arXiv:2604.05222
- P. Zhu, **A. Panigrahi**, L. Levitov, N. Trivedi; *Strain Response as a Probe of Spinons in Quantum Spin Liquids* arXiv:2512.03137
- **A. Panigrahi**, B. Roy; *Projected branes as platforms for crystalline, superconducting, and higher-order topological phases* PRB **113**, 035301 (2026)
- **A. Panigrahi**, V. Poliakov, L. Levitov; *Tomographic imaging of superconducting order using particle-hole interference* PNAS **123** (27) e2534730123 (2026)
- **A. Panigrahi**, A. Kumar; *Non-Fermi Liquids from Subsystem Symmetry Breaking in van der Waals Multilayers* PRL **134**, 236502 (2025)
- **A. Panigrahi**, V. Poliakov, Z. Dong, L. Levitov; *Spin chirality and fermion stirring in topological bands* arXiv:2407.17433
- L. Holleis, T. Xie, S. Xu, H. Zhou, C. L. Patterson, **A. Panigrahi**, T. Taniguchi, K. Watanabe, L. S. Levitov, C. Jin, E. Berg, A. F. Young; *Fluctuating magnetism and Pomeranchuk effect in multilayer graphene* Nature **640**, 355–360 (2025)
- M. Masseroni, M. Gull, **A. Panigrahi**, N. Jacobsen, F. Fischer, C. Tong, J. D. Gerber, M. Niese, T. Taniguchi, K. Watanabe, L. Levitov, T. Ihn, K. Ensslin, H. Duprez; *Spin-orbit proximity in MoS_2/bilayer graphene heterostructures* Nat Commun **15**, 9251 (2024)
- **A. Panigrahi**, L. Levitov; *Signatures of electronic ordering in transport in graphene flat bands* PRB **110**, 035122 (2024)
- **A. Panigrahi**, V. Juričić, B. Roy; *Projected Topological Branes* Commun Phys **5**, 230 (2022)
- **A. Panigrahi**, S. Mukerjee; *Energy magnetization and transport in systems with a non-zero Berry curvature in a magnetic field* SciPost Phys. Core 6, 052 (2023)
- **A. Panigrahi**, R. Moessner, B. Roy; *Non-Hermitian dislocation modes: Stability and melting across exceptional points* PRB **106**, L041302 (2022)

Manuscripts in preparation

EXPECTED TIME OF COMPLETION: SEPTEMBER—OCTOBER 2026

- **A. Panigrahi**, V. Poliakov, L. Levitov; *Effects of two-impurity interference in superconductors with a van-Hove singularity* (final calculation)
- V. Poliakov, **A. Panigrahi**; *Fork-tip Josephson STM as a direct probe of topological superconductivity* (draft ready)
- **A. Panigrahi**, B. Roy; *Topological phase transitions in dilute projected amorphous branes* (draft ready)
- **A. Panigrahi**, B. Roy; *Dynamical phase transitions in interacting relativistic quantum materials* (final calculations)

Research Experience

Effects of Berry curvature in collective modes in chiral superconductors — *MIT, Cambridge, MA, USA*

WITH PROF. CHUNLI HUANG (2026)

- Working on a self-consistent microscopic theory of localized wavepackets and collective modes in a topological superconductor.

Deducing symmetry properties and topological phase winding of unconventional superconductors with STM

MIT, Cambridge, MA, USA

WITH PROF. LEONID LEVITOV AND VLADISLAV POLIAKOV (2024 — PRESENT)

- Developed a theory of particle-hole interference near a pair of impurities to probe the symmetries and topological phase winding of order parameters of unconventional superconductors
- Extended the proposal to realistic unconventional superconductors such as Strontium Ruthenate
- Developed similar STM probes for detecting topological superconductivity with coupled superconducting STM tips and with spin-sensitive STM measuring the spin-chirality of the system.

Strain induced gauge fields as a probe of quantum spin liquids

MIT, Cambridge, MA, USA

WITH PROF. NANDINI TRIVEDI, PROF. LEONID LEVITOV, AND PENGHAO ZHU (2024 — PRESENT)

- Developed a controlled method for calculating strain susceptibility of Kitaev Spin Liquids (KSL) and Chiral Spin Liquids (CSL), as a part of a project establishing that the strain response can be utilized to distinguish KSL, CSL and Intermediate Gapless Phases.

Marginal Fermi liquids resulting from subsystem symmetry breaking

MIT, Cambridge, MA, USA

WITH AJESH KUMAR (2024)

- Demonstrated that subsystem symmetry breaking in van der Waals heterostructures can give rise to an anisotropic marginal Fermi liquid, with quasiparticle lifetime $\tau \sim \frac{1}{|\omega| \log |1/\omega|}$ and specific heat $C \sim T[\log(1/T)]^2$.
- Developed and implemented a Hartree-Fock calculation to determine the parameter regime in which the anisotropic marginal Fermi liquid can arise in realistic systems.

Transport in isospin-ordered phases in graphene

MIT, Cambridge, MA, USA

WITH PROF. LEONID LEVITOV (2023 — 2024)

- Predicted that momentum-polarized nematic phases in biased bilayer graphene can lead to resistance decreasing with rising temperature
- Demonstrated hysteresis-like switching behavior under the action of a strong electric field

Topological phases in projected lower dimensional branes

MPIPKS, Dresden, Germany (remotely)

JOINTLY WITH PROF. BITAN ROY AND PROF. VLADIMIR JURIČIĆ (JUNE 2021 - SEPTEMBER 2021)

- Showed the existence of dislocation modes, Weyl points, and Landau levels in lower-dimensional projected crystals and Fibonacci quasicrystals
- Proposed how this method can be utilized to study higher dimensional (>3D) topological phases within 3D systems

MIT, Cambridge, MA, USA

WITH PROF. BITAN ROY (2025 — PRESENT)

- Established that projected branes can serve as a platform for hosting crystalline, superconducting, and higher-order topological phases.
- Found a new class of quantum phase transition in dilute projected amorphous branes that is distinct from disorder induced transitions in quantum Hall systems.

Research Interests

Broadly interested in theoretical Condensed Matter Physics

- **Tunneling phenomena in unconventional superconductors and in superconductors with broken time-reversal symmetry**
- **Tunneling phenomena in Intervalley Coherent Phases and Altermagnets**
- **Electronic transport in two-dimensional systems and the effects of Berry curvature in transport**
- **Non-Fermi Liquids emerging due to subsystem symmetry breaking**
- **Probes for Quantum Spin Liquids**
- **Topological phases of matter and Quantum Phase transitions**

Skills

Programming skills	Julia, MATLAB/Octave, Mathematica, Python
Advanced Physics Courses	Strongly Correlated Systems, Advanced Statistical Physics, Quantum Field Theory I, General Relativity
Languages	Fluent in English, Bengali, Hindi

Talks

Imaging Superconducting Order with STM via Particle-Hole Interference (Invited talk)

Maglab CMS Journal Club, Tallahassee, Florida, USA (remotely)

CLICK HERE TO DOWNLOAD THE PRESENTATION

August 2026

Inducing chiral bound states in multilayer graphene via Berry curvature: analytical and numerical approach

APS March Meeting, Denver, CO

CLICK HERE TO DOWNLOAD THE PRESENTATION

March 2026

Strain response as a probe of Spinon bands in Quantum Spin Liquids — *APS March Meeting, Denver, CO*
CLICK HERE TO DOWNLOAD THE PRESENTATION — *March 2026*

Tunneling density of states in exotic superconductors and spatial patterns of particle-hole interference — *APS March Meeting, Anaheim, CA*
CLICK HERE TO DOWNLOAD THE PRESENTATION — *March 2025*

Transport Signatures of Electronic Ordering in Graphene Flat Bands — *Indian Institute of Science, Bangalore, India*
CLICK HERE TO DOWNLOAD THE PRESENTATION — *January 2024*

Topological phases in quasicrystals: A general principle of construction — *APS March Meeting (virtually)*
CLICK HERE TO DOWNLOAD THE PRESENTATION — *March 2022*

Dislocation as a bulk probe of non-Hermitian topology — *MPIPKS, Dresden, Germany (remotely)*
CLICK HERE TO DOWNLOAD THE PRESENTATION — *July 6, 2021*

Teaching Experience

Theory of Solids II (8.512) — *MIT*
TEACHING ASSISTANT — *Spring 2026*

- Conducted recitation classes on various topics in condensed matter physics

Physics II: Electricity and Magnetism (8.102) — *MIT*
TEACHING ASSISTANT — *Spring 2024*

- Taught students one-on-one in office hours and graded exams

Academic Achievements

2022	**1st Rank in India** in CSIR-NET (JRF) 2021 in Physics, held in February 2022 due to COVID (score 186/200)	*India*
2022	**1st Rank in India** in Graduate Aptitude Test in Engineering (G.A.T.E.) in Physics	*India*
2017-22	CGPA **9.8/10** in B.S. (Research) and M.S., received **Prof. R. Srinivasan Medal** for **highest CGPA** in batch	*IISc, Bangalore*
2017	**1st rank (99.2 %) in Board** in Higher Secondary Examination, among about 0.7 million candidates	*West Bengal, India*
2015	**2nd rank (97.57 %) in Board** in Secondary Examination, among about 1 million candidates	*West Bengal, India*

References

- Prof. **Leonid Levitov**, Dept. of Physics, Massachusetts Institute of Technology, Cambridge, MA 02139, USA.
  Email Address - levitov@mit.edu
- Prof. **Bitan Roy**, Dept. of Physics, Lehigh University, Bethlehem, PA 18015, USA.
  Email Address - bitan.roy@lehigh.edu
- Prof. **Nandini Trivedi**, Dept. of Physics, The Ohio State University, Columbus, OH 43210, USA.
  Email Address - trivedi.15@osu.edu
- Prof. **Chunli Huang**, Dept. of Physics and Astronomy, University of Kentucky, Lexington, KY 40506, USA.
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